

Math 111: Calculus Practice Exam 2

For Exam 2, in class on Friday, October 16

What Exam 2 is like. Four problems in a fifty-minute period, closed-book and device-free. It is **cumulative**: the new material is Chapter 2 (the derivative rules, including the chain rule and inverse functions), §3.1 (related rates), and §3.3 (extreme values), but ideas from Unit 1 — reading f' from a graph or a table, over- and under-estimates, the tangent line — come back inside the new problems. Each problem is marked holistically on the 0–5 scale: a right answer with no reasoning is not a 5, and a wrong answer containing a real idea is not a 0. The derivative rules are now fair game and expected; say which rule you are using and why it applies.

How to use this. Sit down with a blank sheet and fifty minutes on a clock, book closed and phone in another room. Bring whatever you could not finish to review on Wednesday the 14th. Problem 4 uses §3.3, which we cover on Monday the 12th.

1. Structure first

about 10 minutes

For each function, first say in a few words how it is built (a sum? a product? a quotient? a composition, and of what?), then differentiate it. You need not simplify.

(a) $f(x) = x^2 \cos x$

(b) $g(x) = \frac{e^x}{x+1}$

(c) $h(x) = \sqrt{1 + \sin(3x)}$

(d) $k(x) = \ln(1 + x^2)$

(e) Find the tangent line to the graph of f at $x = \pi$, and use it to estimate $f(3)$. (You may leave π in your answer.)

2. A balloon, a barometer, and two tables

about 13 minutes

A weather balloon rises. Its altitude $a(t)$, in meters, is recorded every five minutes:

t (minutes)	0	5	10	15	20
$a(t)$ (m)	0	400	750	1050	1300

Air pressure $P(a)$, in kilopascals, depends only on altitude a :

a (m)	0	500	1000	1500	2000
$P(a)$ (kPa)	101.3	95.5	90.0	84.8	79.9

The pressure the balloon experiences at time t is $H(t) = P(a(t))$. There are no formulas; the tables are all you have.

(a) Estimate $a'(10)$, with units. Then give one estimate of $a'(10)$ that is certainly too large and one that is certainly too small, and say what you are assuming about a' in order to be sure.

(b) Estimate $P'(750)$, with units. Say in one sentence what the sign of this number means physically.

- (c) Estimate $H'(10)$, with units. Explain why the two numbers you combine should be combined the way you combined them — in terms of what happens to the balloon over the next minute, not by naming a rule.
- (d) A classmate computes $H'(10)$ as $P'(10) \cdot a'(10)$. Say precisely what is wrong, and why the number $P'(10)$ is not even the kind of quantity that belongs here.

3. A melting snowball

about 12 minutes

A snowball is melting. Assume it stays a perfect sphere, and that its volume is decreasing at a steady 2 cubic centimeters per minute. Recall $V = \frac{4}{3}\pi r^3$ and surface area $S = 4\pi r^2$.

- (a) Draw and label a picture, and write down, with units, every rate you know and the rate you want.
- (b) How fast is the radius shrinking at the moment the radius is 3 cm?
- (c) How fast is the surface area shrinking at that moment?
- (d) Is the radius shrinking faster or slower when the snowball is smaller? Answer from your formula in (b), then say in a sentence why that makes physical sense.

4. All about f'

about 13 minutes

A function f is differentiable everywhere, and its derivative is

$$f'(x) = (x - 2)(x + 1)^2.$$

You are not told f itself.

- (a) Find all critical numbers of f .
- (b) Make a sign chart for f' , and use the first derivative test to classify each critical number as a local maximum, a local minimum, or neither. Justify each from the chart.
- (c) Compute f'' and apply the second derivative test at each critical number. Where the test is silent, say so, and say which test settles the matter.
- (d) A classmate writes: “ $f'(-1) = 0$, and f' is negative on both sides of -1 , so the graph of f turns around at $x = -1$ and f has a local maximum there.” Find the first wrong step, say why it is wrong, and describe what the graph of f actually does near $x = -1$.
- (e) On axes of your own, sketch a possible graph of f on $[-3, 4]$, consistent with everything above. Mark the critical points.